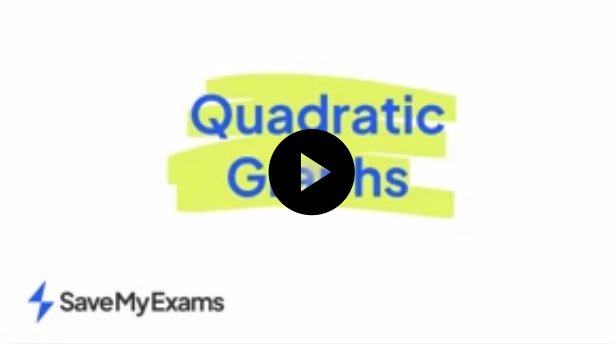
Quadratic Graphs

Exam code: 9MA0

Download PDF

Written by: Paul
Updated on 19 June 2025

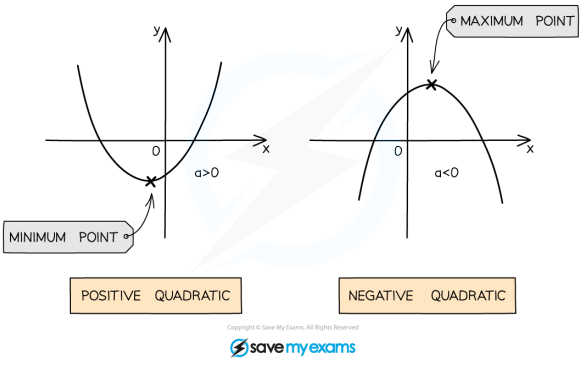


Did this video help you? Yes No

Quadratic graphs

What are quadratic graphs?

- The general equation of a quadratic graph is $y = ax^2 + bx + c$
- Their shape is called a parabola ("U" shape)
- Positive quadratics have a value of $a > 0$ so the parabola is upright U
- Negative quadratics have a value of $a < 0$ so the parabola is upside down ∩



How do I use quadratic graphs?

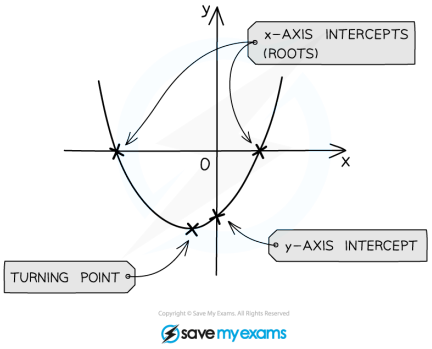
You need to be able to:

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- **factorise**, if possible, to find the roots of the quadratic function
- find the coordinates of the **turning point** (maximum or minimum)

You may have to rearrange the equation before you can find some of these things



Examiner Tips and Tricks

- Your calculator may tell you the **roots** of a quadratic function and the coordinates of the turning point
- But don't rely on it – think about how many marks the question is worth and how much method/working you should show
- Remember sometimes you'll need to rearrange an equation into the form $y = ax^2 + bx + c$

Worked Example

of the axes intercepts and the turning point.

$y = 2x^2 + 5x - 12$ — POSITIVE QUADRATIC: U-SHAPED

AT $x = 0$, $y = -12$ — SET $x = 0$ FOR y-AXIS INTERCEPT

$y = (2x-3)(x+4)$ — FACTORISE AND SOLVE FOR x-AXIS INTERCEPTS

$y = 0$ AT $x = \frac{3}{2}$ AND $x = -4$

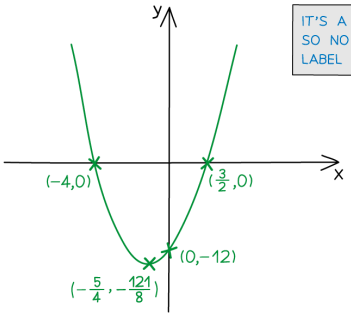
$\therefore$ AXES INTERCEPTS ARE $(0,-12)$, $(-4,0)$ AND $(\frac{3}{2},0)$

$y = 2(x^2 + \frac{5}{2}x) - 12$ — COMPLETE THE SQUARE...

$y = 2[(x+\frac{5}{4})^2 - \frac{25}{16}] - 12$

$y = 2(x+\frac{5}{4})^2 - \frac{121}{8}$ — ... TO FIND THE TURNING POINT

$\therefore$ TURNING POINT IS $(-\frac{5}{4}, -\frac{121}{8})$



IT'S A SKETCH SO NO NEED TO LABEL AXES

Teacher only

New! Exclusive Teacher-Only Questions

- Fresh, expert-written questions
- Students can't access them
- Fair accurate tests - built fast

Explore in Test Builder

Test yourself

Was this revision note helpful? Yes No

Resources

Learning Hub

Scholarship

Join

Past Papers

Solution Banks

Sitemap

Dashboard

Overview

Brand ambassador

Profile

Subscription

Settings

Log out

Company

About us

Exam Specificity

Content Quality

Promotions

Jobs

Terms

Privacy

Cookie Policy

Help and Support

Subjects

Biology

Chemistry

Physics

Maths

Geography

English Literature

Psychology

All Subjects

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.

Neither IBO nor College Board were involved in the production of, and do not endorse, the resources created by Save My Exams.